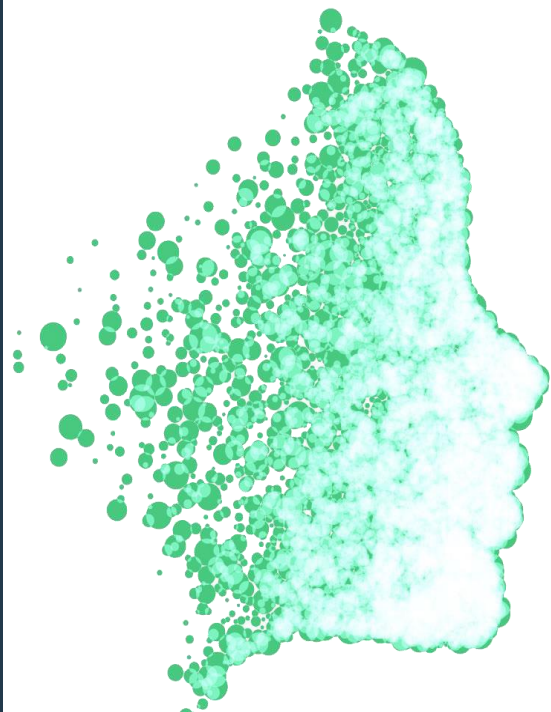




I have a dream



Connected and Autonomous Vehicles

PhD course – part 2

© 2026 HiPeRT Lab



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

High Performance
Real Time **Lab**



Course outline

Lesson #1

- › Intro course + basics of AD
- › Perception and AI
- › Hardware platforms

Lesson #2

- › Cars and the city
- › Navigation: planning and control
- › Let's build the car

Assignments





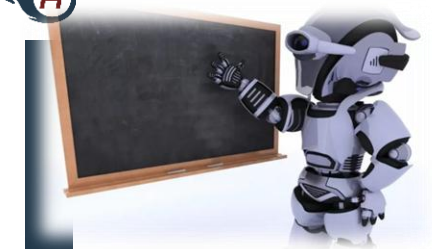
Autonomous driving

&

Smart cities



1. Train our AI



2. Additional eyes
for the car!!



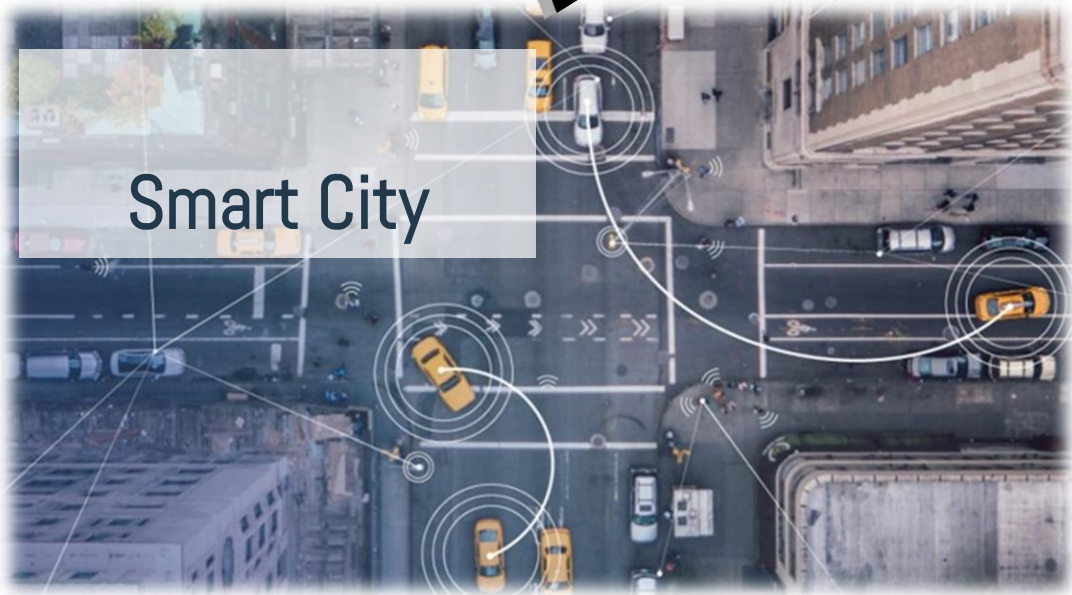


- ✓ Raw data for training sets
- ✓ Privacy and storage issues
- ✓ Real-time processing

1. Train our AI



2. Additional eyes for the car!!



- ✓ City cameras, other vehicles, pedestrians, traffic lights, traffic signs....
- ✓ Internet-of-Vehicles
- ✓ Stable 4-5G network

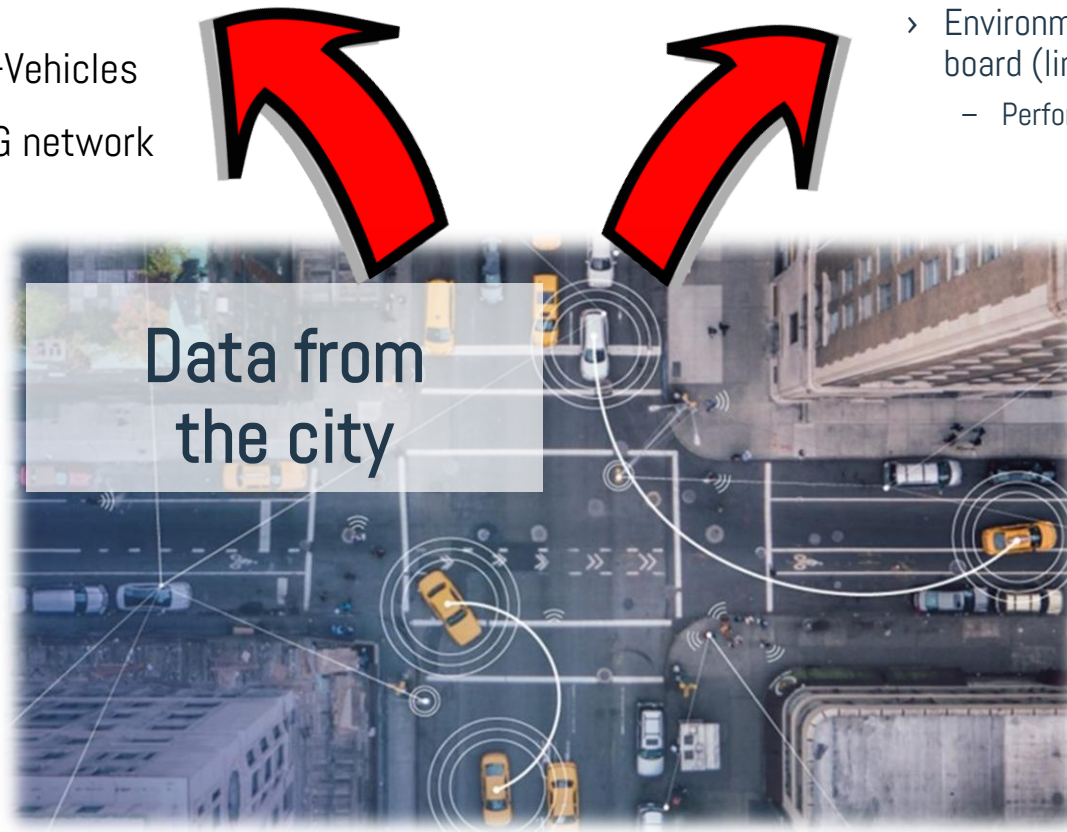


City IoT

- ✓ City cameras, other vehicles, pedestrians, traffic lights, traffic signs....
- ✓ Internet-of-Vehicles
- ✓ Stable 4-5G network

We use

- › Raw data for training sets
 - Privacy and storage issues
- › Environment information to enhance on-board (limited) perception
 - Performance / real time



HOME

COS'È MASA ▾

SERVIZI ▾

COMUNICAZIONE ▾

CONTATTI

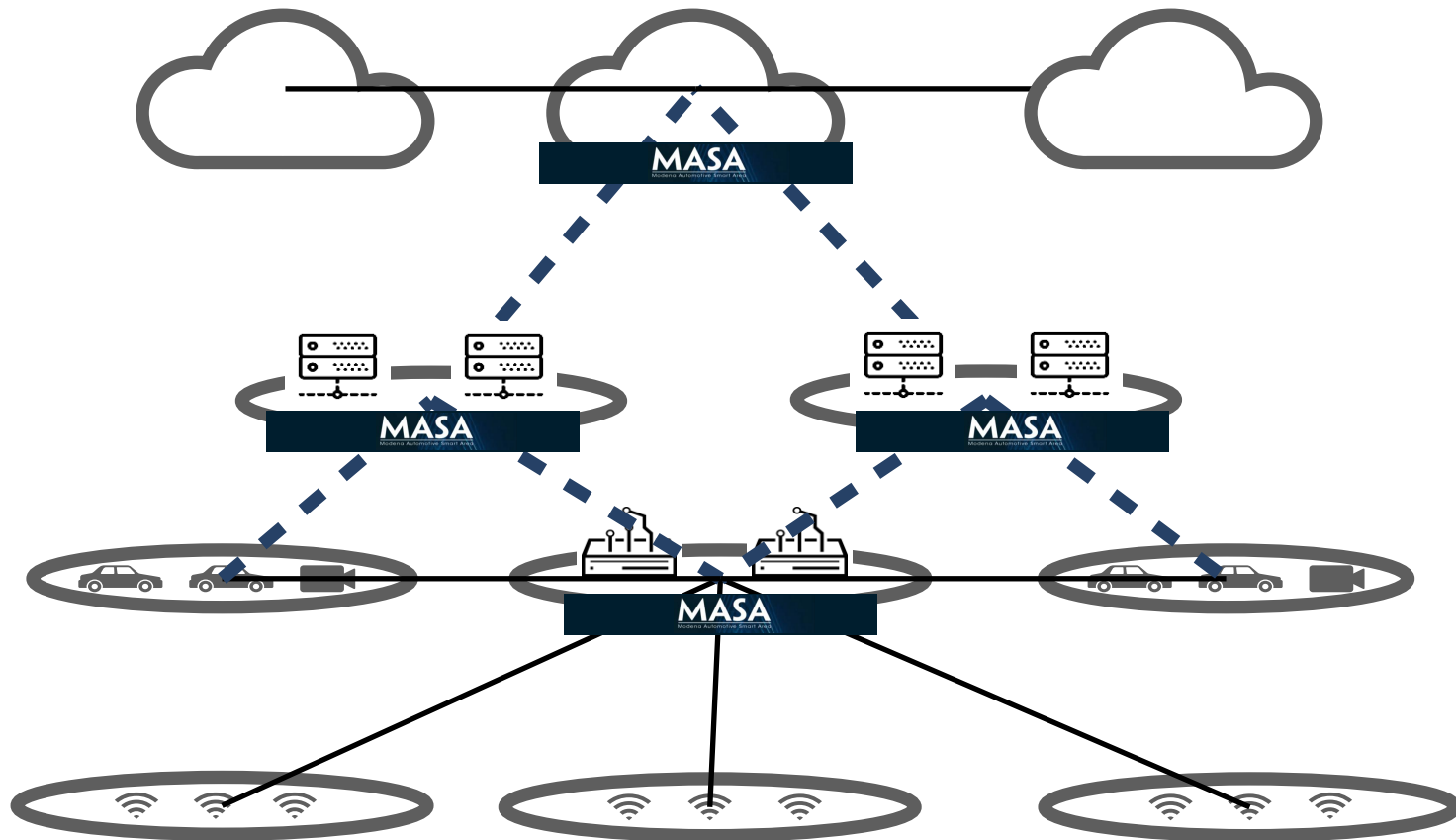
🇮🇹 ▾

MASA

Modena Automotive Smart Area

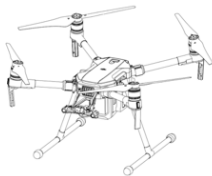
A LIVING LAB FOR AUTOMATED DRIVING

MASA è il primo laboratorio "a cielo aperto" per la sperimentazione e la certificazione delle tecnologie di guida autonoma e di guida connessa (veicoli, componenti e servizi) che interagiscono con altri veicoli, con le infrastrutture e con gli elementi dello spazio urbano.





What, where, how?



On field (aka: on-edge)

- › Perception; log and analyze
- › Short term sense, predict & react in RT



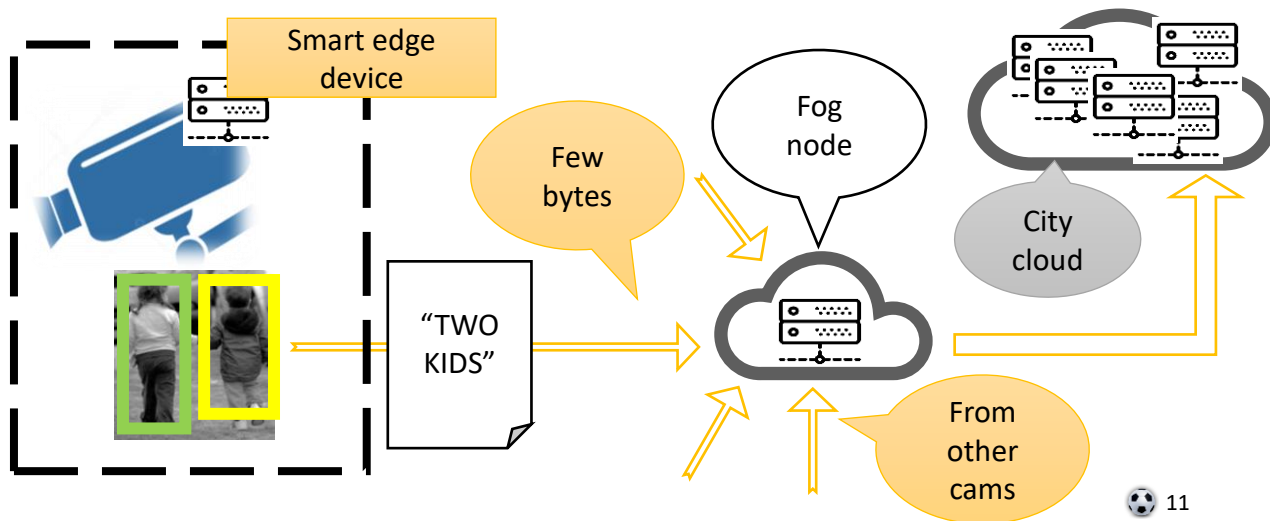
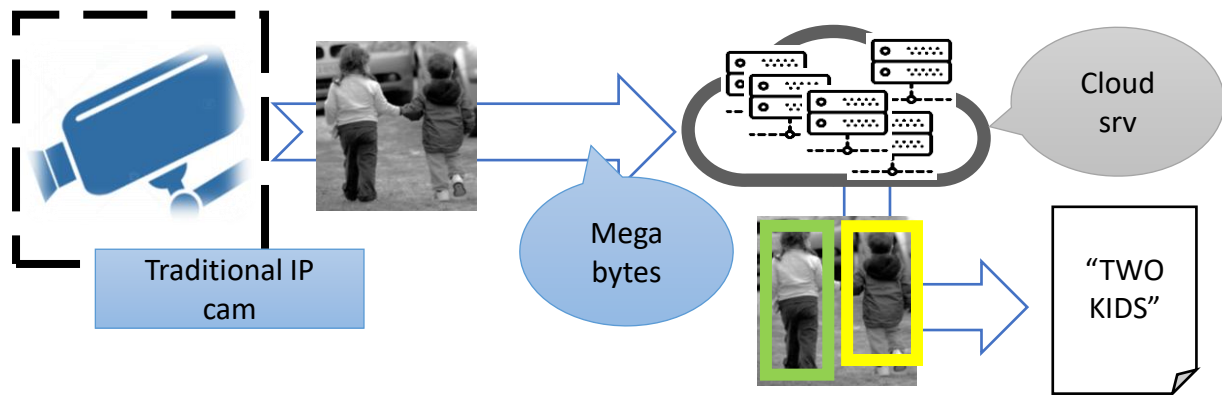
In cloud

- › Aggregate perception from multiple edges
- › Big data analytics and logging
- › Long-term “holistic” prediction



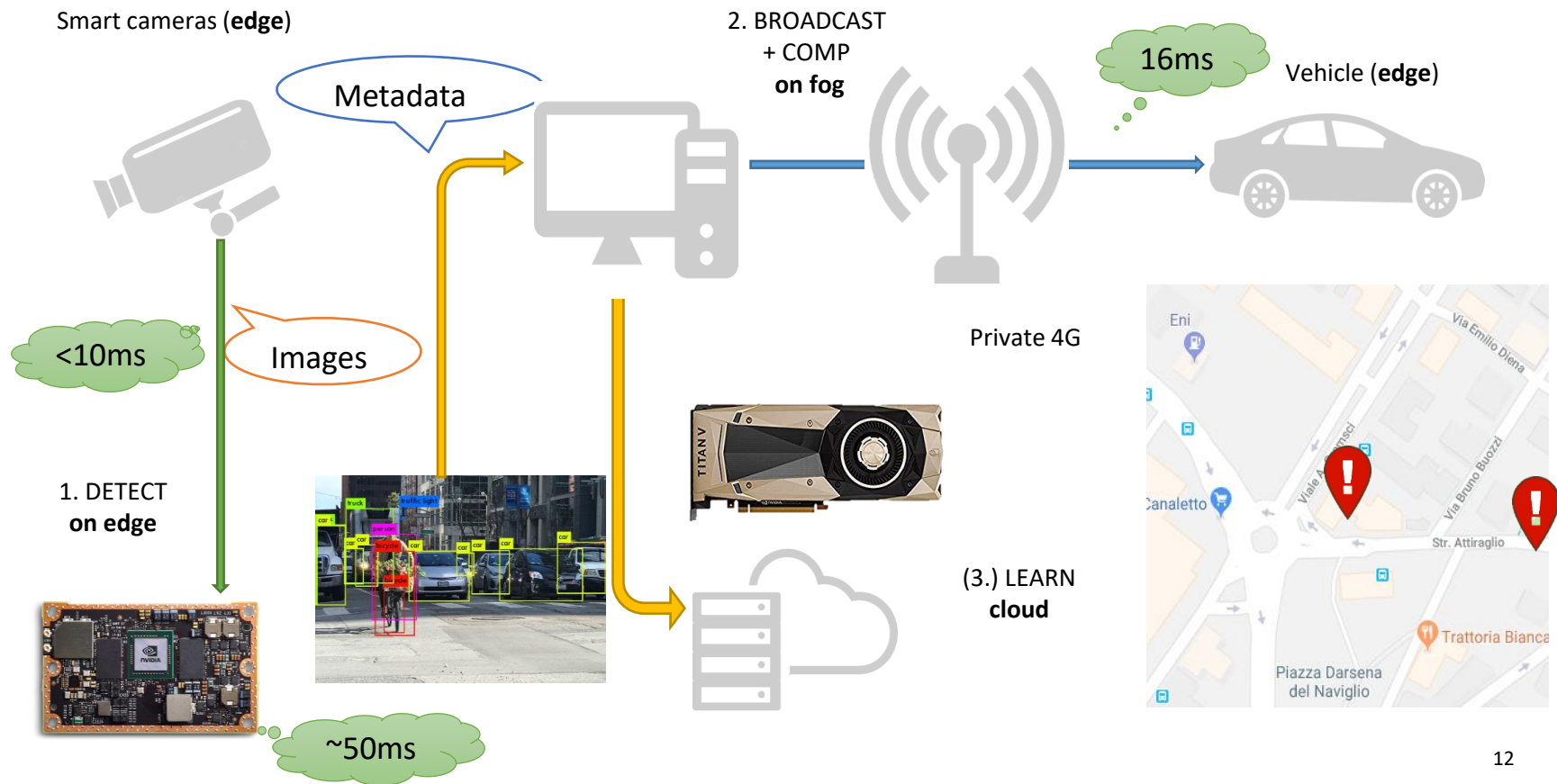
Comms

- › Safe, reliable and secure





Fast Smart City flow: under 100 *ms*





Cars speak among them

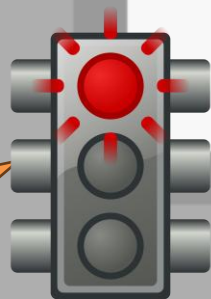


Cars speak to/from the city





-3 seconds
to green

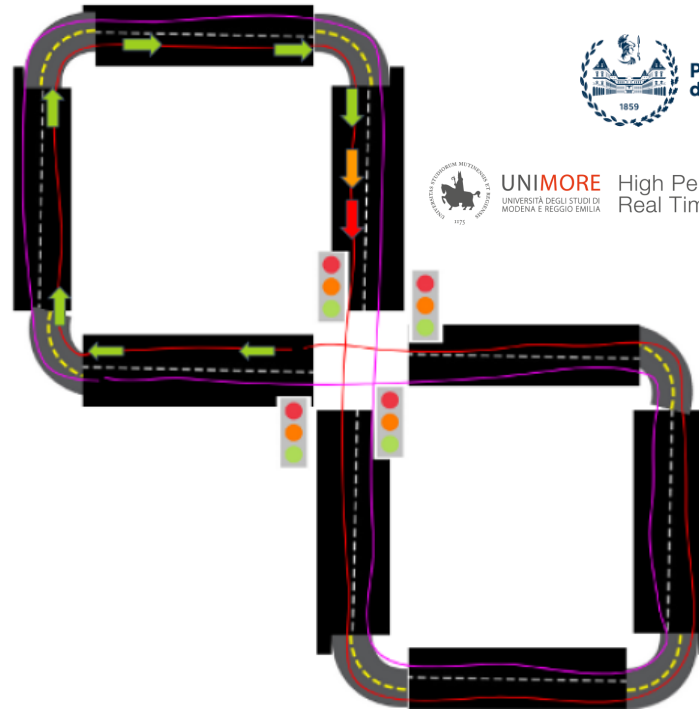




The MoVeOver project

- › Funded by the PNRR (Funding for the Italian resiliency)
- › Target: smart intersections for Connected Autonomous Vehicles
- › Traditionally prototyped in simulators / in Matlab...

Now, let's create it!



UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA



Politecnico
di Torino



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

High Performance
Real Time Lab



Finanziato
dall'Unione europea
NextGenerationEU



Ministero
dell'Università
e della Ricerca



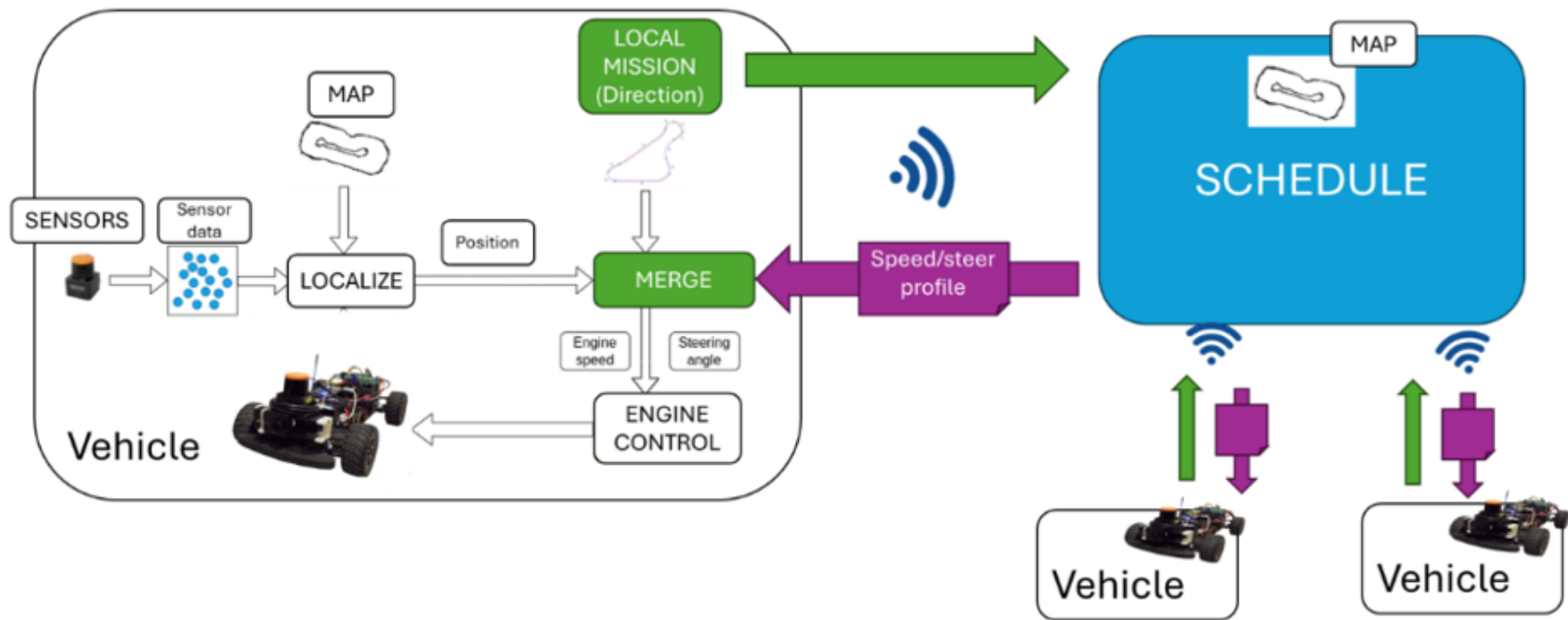
Italiadomani
PIANO NAZIONALE
DI RIPRESA E RESILIENZA

RESTART



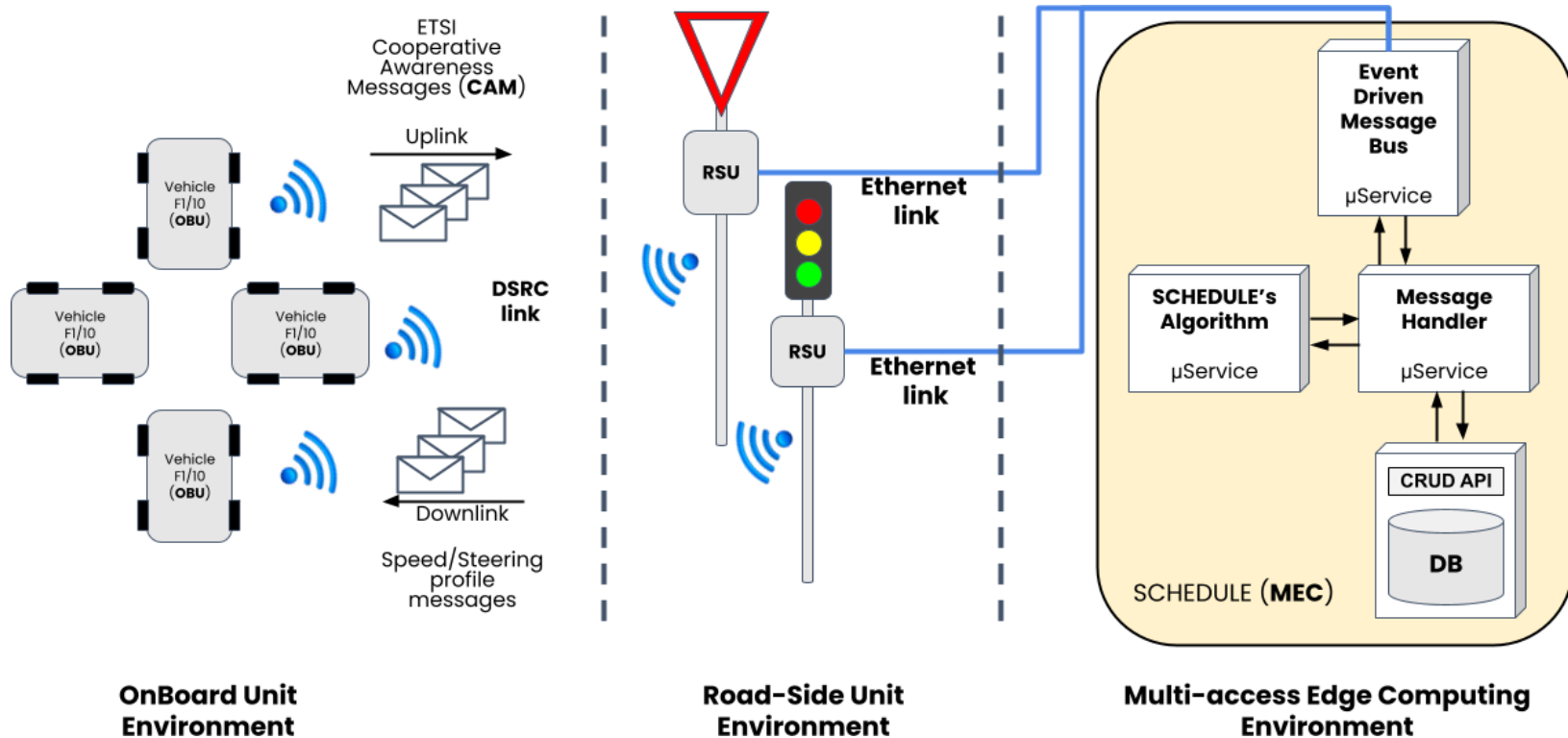
System architecture

- › The decision module internally we call "SCHEDULE"

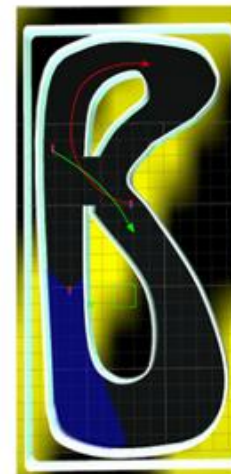
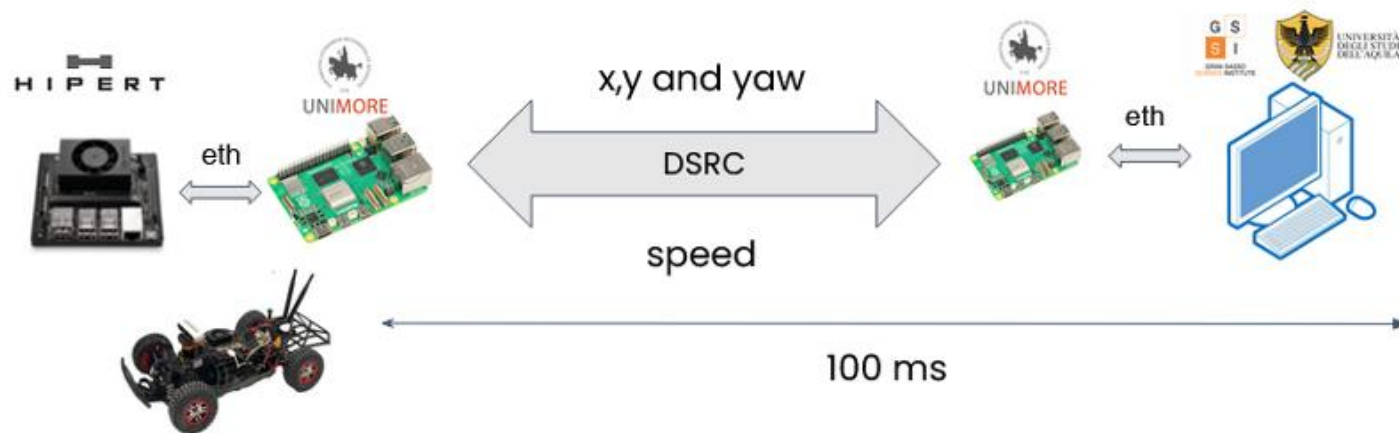




Communication infrastructure

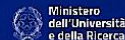


Communication infrastructure – on-vehicle





Not only a toy...



RESTART Final Plenary Dissemination Event

"Shaping Horizons In Future Telecommunications" | January 19-21, 2026 | Rome

THE NEW APPLICATIONS | BEST DEMO
F45.1 Mini-cars demonstration of
project traffic control
algorithms

RESTART Scientific Coordinator
Prof. Antonio Capone

RESTART Program Coordinator
Prof. Nicola Blefari Melazzi

"Shaping Horizons In Future Telecommunications" | January 19-21, 2026 | Rome





How are cars built?

Next-generation ECUs



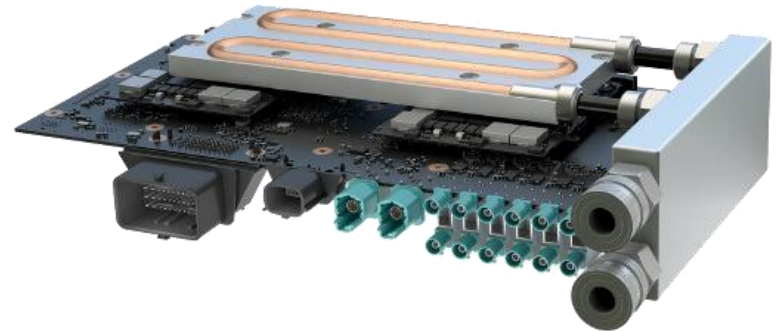
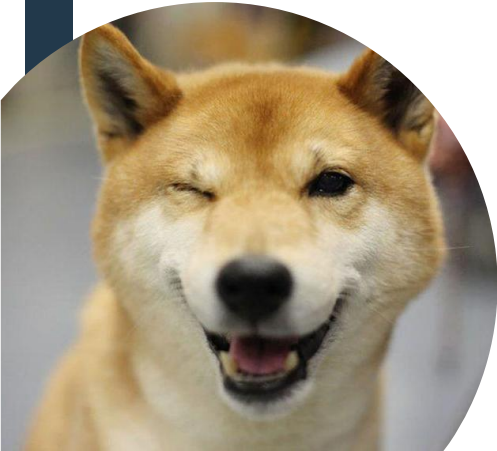
"What's the problem?"





A new generation of computers

High-Performance embedded systems (est. 2014-16)

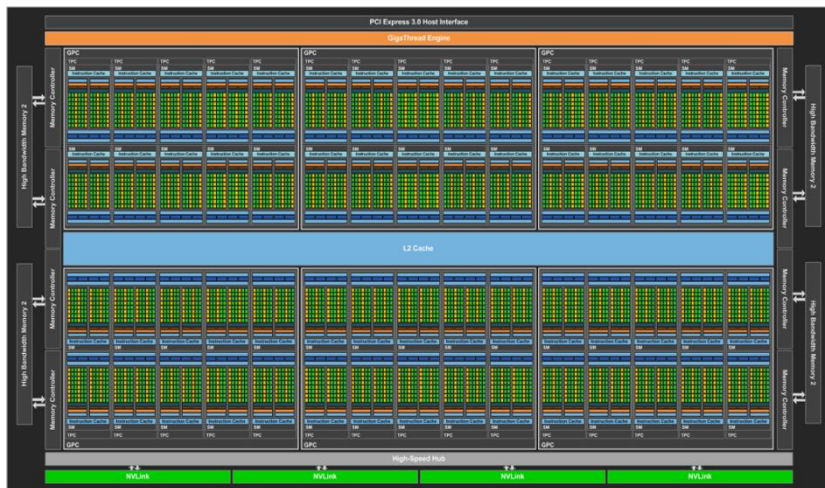




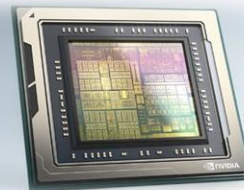
Heterogeneous computing architectures



CPU
MULTIPLE CORES



GPU
THOUSANDS OF CORES



E.g.: Nvidia Orin
200 TOPS at 40W



IBM Blue Gene (2007): 131,000 CPUs, 250 m², 1+MW (+cooling)



Heterogeneous computing architectures

Couple a General-Purpose Host....

- › E.g., ARM, HiSilicon
- › Typically run GNU/Linux

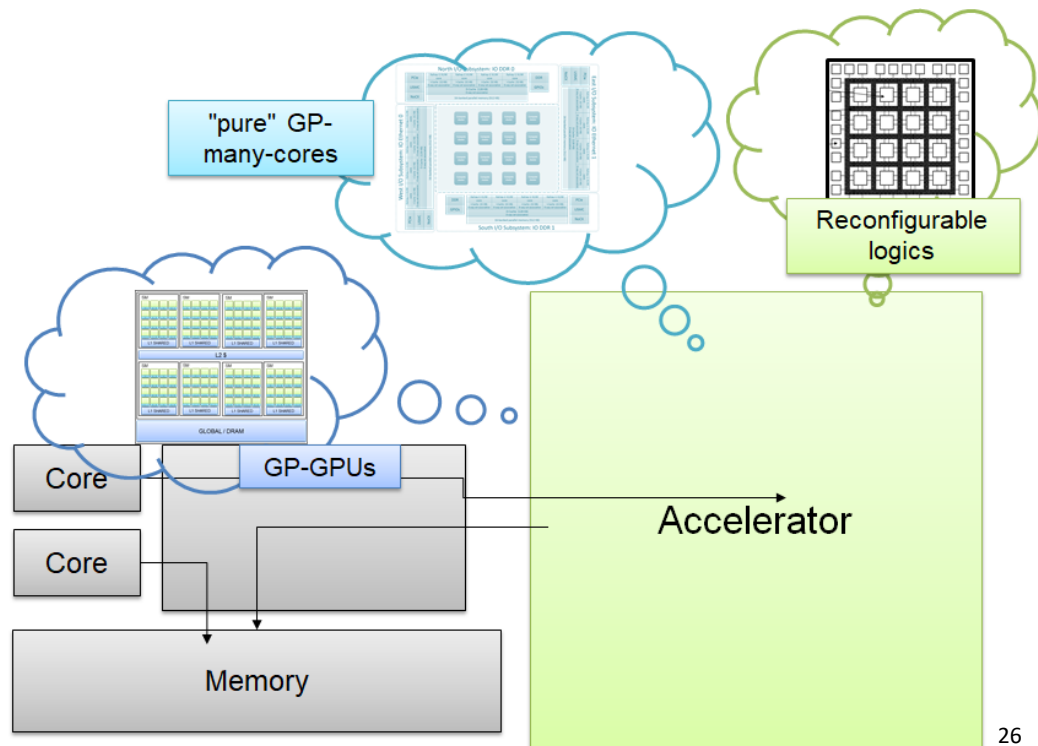
...and 1+ data cruncher accelerator(s)

- › GPGPUs vs FPGAs vs ASIC vs TPUs vs DLEs vs....

Key requirements

- › < 30/50W
- › > 2TFLOPS
- › (T)FLOPS/W is the metric

SWaP – Size, Weight and Power





One step back...



Neural networks

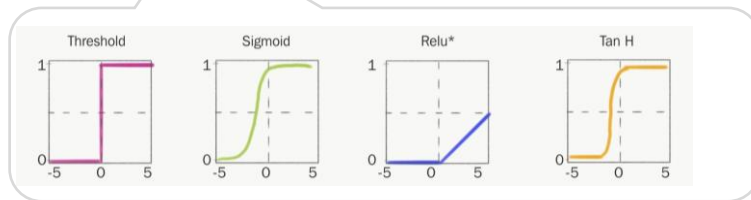
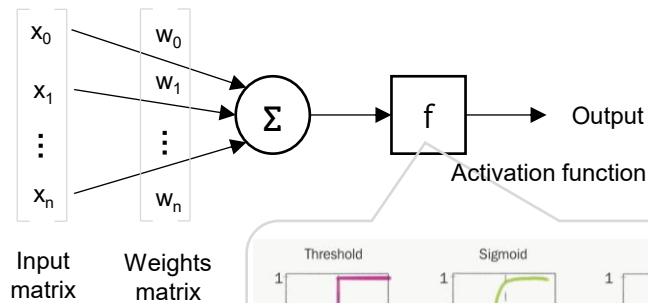
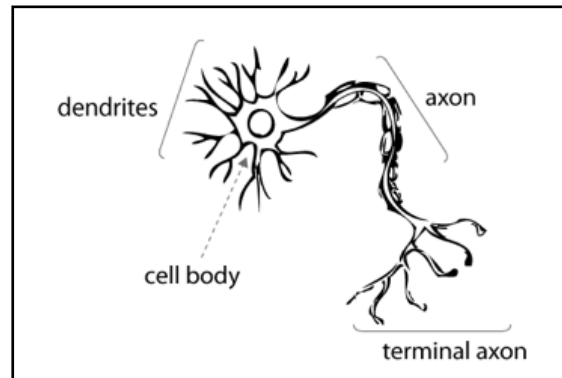
Bio-inspired

Based on neurons arranged in layers

- › And sub-layers

Convolutional neural network

- › Convolutions are widely used in computer graphics (keep this in mind..)

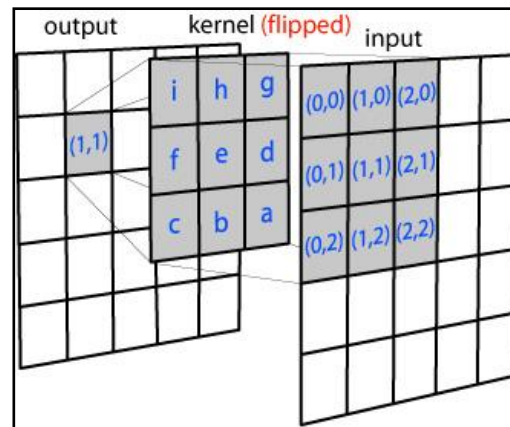




Convolutions

- › Computation-intensive
- › Suitable for implementation in hardware
- › In computer vision, blurring

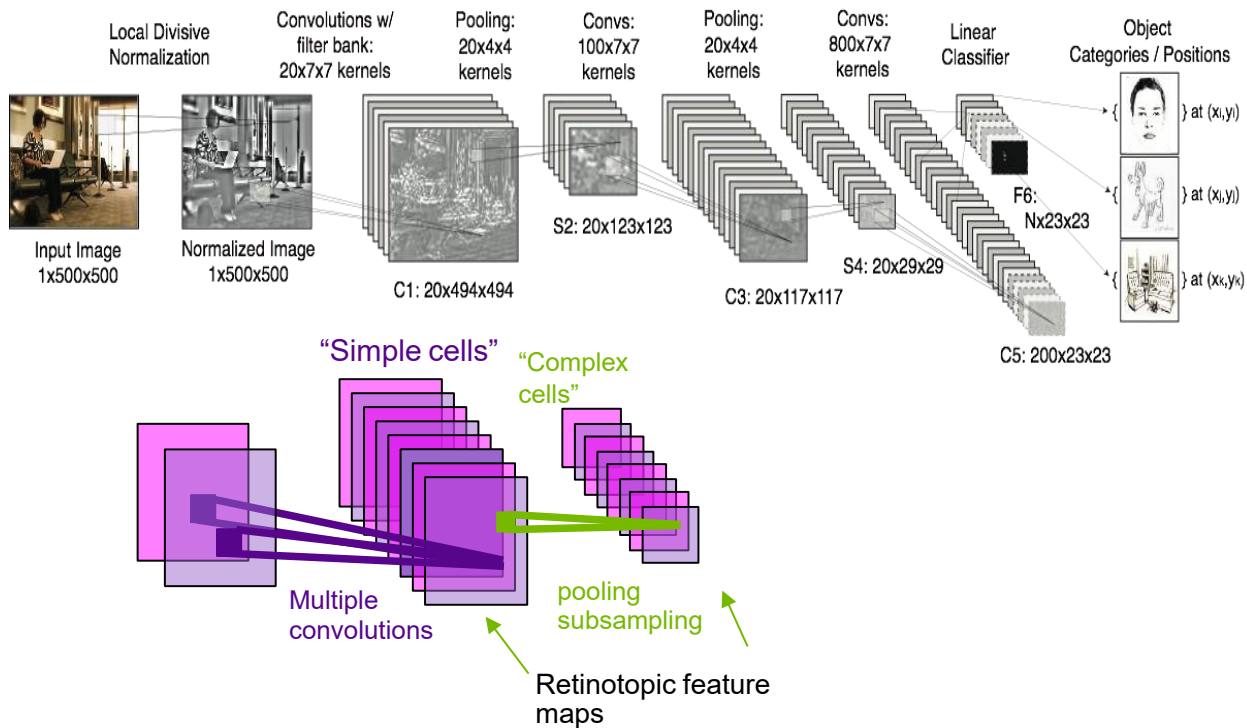
$$\begin{aligned}(f * g)(t) &\stackrel{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} f(t - \tau)g(\tau) d\tau.\end{aligned}$$





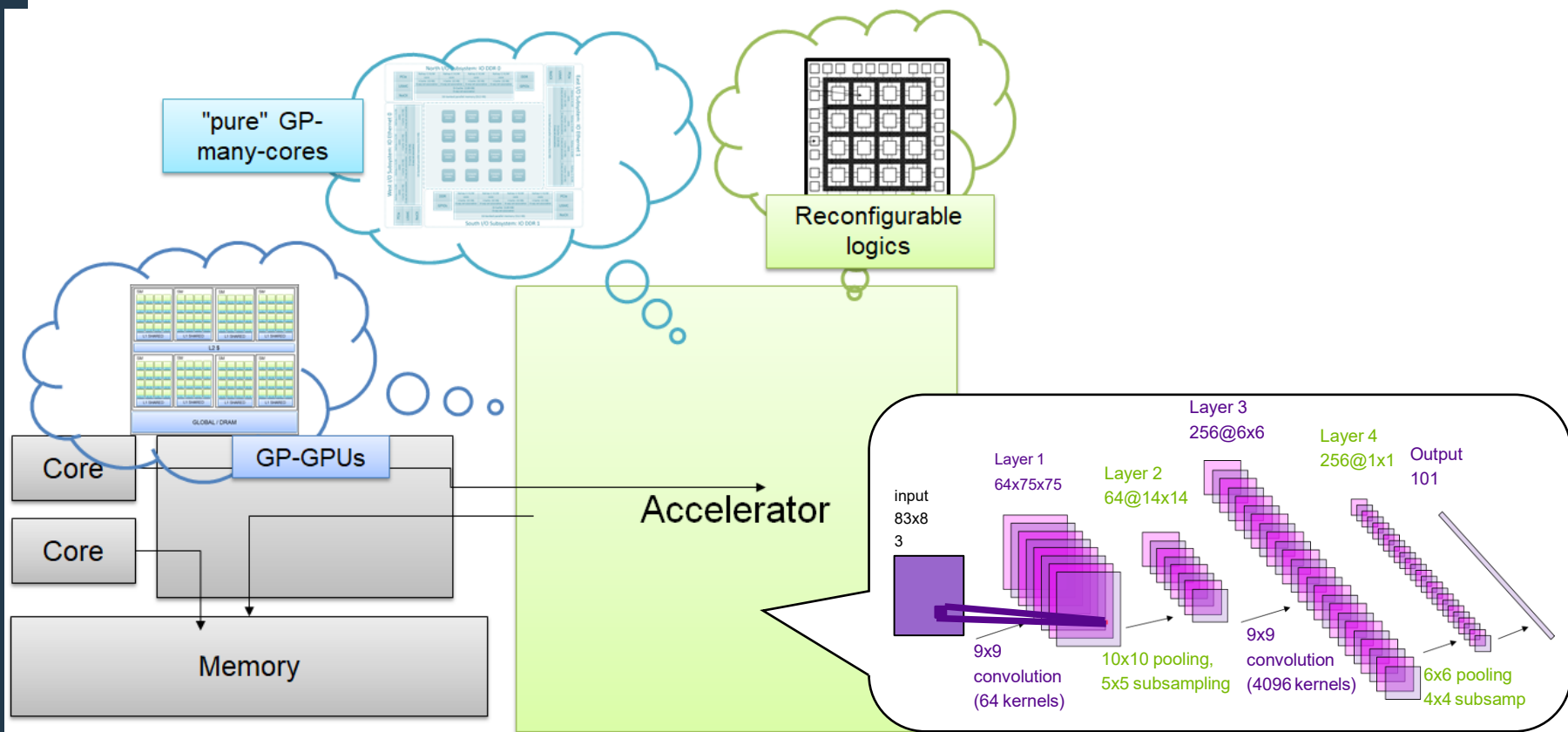
The convolutional net model - CNN

> (Multistage Hubel-Wiesel system)





How to deploy a neural network





Embedded computers

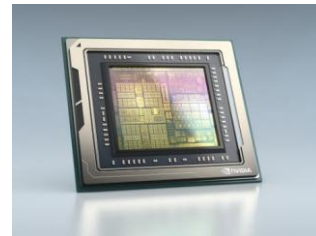
2007:

IBM Blue Gene
200+ TFLOPS

131,000 CPUs

250 m2

1+MW (+ cooling)

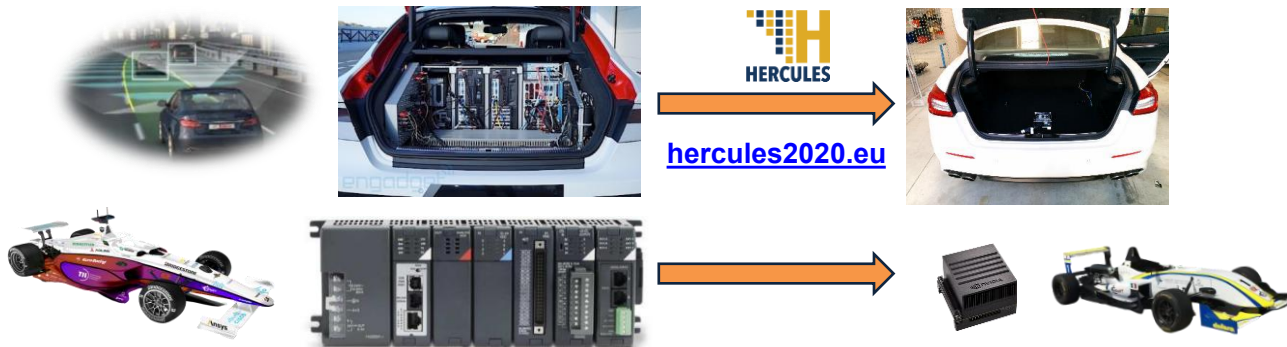


2022:

Nvidia Orin
200+ TOPS

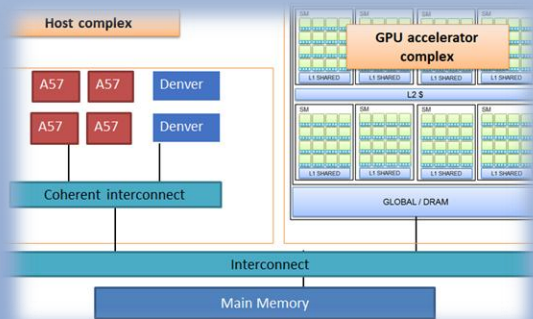
40W

Reduced Size, Weight & Power

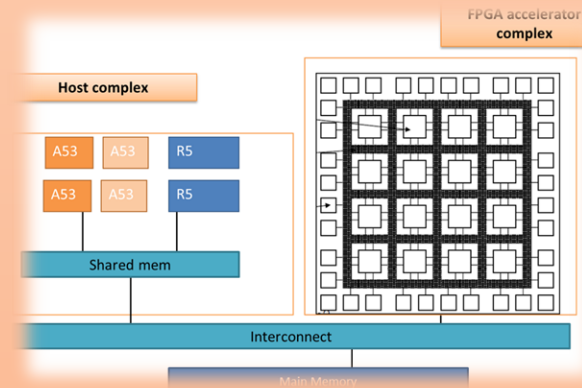




GPGPU SKY HARD



MARVEL
CAPTAIN AMERICA
CIVIL WAR



W LA FPGA



Real-Time Embedded Systems

Cars are constrained!

- › Size, Weight and Power – SWaP
- › High workload (e.g, Machine Learning), require high-performance and power efficiency
- › Example: How does it take to turn on MR-23DL (“Diletta”)?

Safety critical systems require predictable performance

- › At all the level of the software stack
- › Not only application...but also RTOSes, hypervisors...



Three dimensions

Workload-intensive complex tasks

- › Data crunching
- › Perception, AI, but also vehicle models and linear/non linear control
- › Example: How does it take to turn on MR-23DL ("Diletta")?

Latency-critical control tasks

- › Cyber-physical interaction
- › Tight control loops (e.g., steering PID)

Safety critical systems require **predictable performance**

- › To be certifiable
- › Full-stack! From HW to OS to SW...



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

High Performance
Real Time **Lab**



References



Course website

- › <https://github.com/HiPeRT/F1tenth-PhD>

My contacts

- › paolo.burgio@unimore.it
- › <http://hipert.mat.unimore.it/people/paolob/>

Resources

- › <https://f1tenth.org>
- › A "small blog"
 - <http://www.google.com>